ICT251: Database System Design and Management

Rational

Database systems are the backbone of modern information management in both business and technology. With the ever-increasing demand for efficient data storage, retrieval, and analysis, professionals well-versed in database design and management are in high demand across various industries. This course is designed to meet the growing need for individuals who can create, manage, and utilize databases effectively while ensuring data integrity and security.

Learning outcomes

This course aims to equip students with the skills and knowledge necessary to design, develop, and manage databases effectively. They will be well-prepared to ensure data integrity, security, and efficient data retrieval, and to provide essential database administration in both enterprise and emerging online and mobile platforms.

Course Objectives

By the end of the course-tenure the students should be able to:

- Understand the fundamental concepts of database systems, encompassing files, records, fields, databases, DBMS architecture, Relational Algebra, and Structured Query Language (SQL).
- Develop the ability to create and maintain databases by grasping the principles of conceptual data modeling, producing Entity Relationship Diagrams (ERDs), and defining entity types and attributes.
- Comprehend entity relationships, including their introduction, relationship cardinality, and optionality.
- Analyze and handle complex relationships, such as associative, recursive, and exclusive relationships, along with the concept of link entity identifiers and subtypes.
- Master logical database design, including different types of keys, resolving various relationship types, and addressing normalization issues.
- Gain expertise in database normalization, from Un-normalized Form (UNF) to Third Normal Form (3NF), while also understanding denormalization and checking for model correctness.

- Implement Database Management Systems (DBMS) using platforms like MySQL, MS SQL Server, MS Access, or Oracle, which includes installation, database creation, table relations, data manipulation, foreign key usage, and advanced data selection techniques.

Course content

1. Introduction and Concept of database system

- Files, records, fields, and database
- DBMS – Architecture
- Relational Algebra
- Structured Query Language (SQL)
- SQL set of commands
- Data manipulation using SQL examples
- Various SQL joins statements

2. Database development

- Conceptual data modelling
- The Entity Relationship Diagram (ERD)
- Entity types
- Producing the ERD
- Entity attributes
- Entity Attributes Diagram Representation
- Entity selection and validation
- Entity definitions
- Validating the model

3. Entity Relationships

- Introduction
- Relationships
- Relationship cardinality
- Relationship optionality

4. Complex relationships

- Introduction
- Associative relationships
- Link entity identifier
- Recursive relationships
- Sub types
- Exclusive relationships

5. Logical Database Design

- Introduction
- Different types of Keys
- Different types of relations
- Resolving many-to-many relationships
- Resolving one-to-many relationships with optionality
- Resolving one-to-one relationships
- Recursive relationships
- Exclusive relationships
- Identification dependency
- Modelling problems

6. Normalisation

- Introduction
- Un-normalised form (UNF)
- First Normal Form (INF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Denormalization
- Checking the model

7. Introduction to Database Management Systems (DBMS) Implementation

(Use at least any two from either: MySQL, MS SQL Server, MS Access and/or Oracle)

- Installation of a DBMS
- Creating creating databases and tables
    - Relating tables and creating ER Diagrams
- Inserting and updating data in a DBMS
- Using Foreign Keys
- Selecting data from a table
- Selecting data from multiple tables
- Subqueries and group functions
    - Creating pages & reports

Teaching and Learning Methods

Lectures:45Hrs Tutorials:30Hrs, Labs:30Hrs, Assessments:15Hrs, NATIONAL HRS: 120

Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 20% for hands-on assessments
  - 30% for theory assessments

Recommended Readings

- Coronel, C., Morris, S., & Rob, P. (2020). Database Systems: Design, Implementation, and Management. Cengage Learning.
- Elmasri, R., & Navathe, S. B. (2019). Fundamentals of Database Systems. Pearson.
- Hernandez, M. J. (2013). Database Design for Mere Mortals: A Hands-On Guide to Relational Database Design. Addison-Wesley.
- Taylor, A. G. (2020). SQL For Dummies. For Dummies.
- Sipser, M. (2012). Introduction to the Theory of Computation. Cengage Learning.
- Beaulieu, A. (2009). Learning SQL: Generate, Manipulate, and Retrieve Data. O'Reilly Media.